DATA 202 — Quiz 3

ANSWER KEY — Version B

Question 1 (SLO 04A)

Table	Column	Answer
`checkins`	`member_id`	Foreign key -> members (it repeats: one row per member per month)
`plans`	`plan`	Primary key
`members`	`calories`	Neither (a measurement, not an identifier)
`members`	`plan`	Foreign key -> plans
`members`	`member_id`	Primary key

Question 2 (SLO 04B)

how=	rows
`"inner"`	5
`"left"`	7
`"right"`	7
`"outer"`	9

M006 and M008 have 2 check-in rows each and M010 has 1, all matching a member, so inner = 5. Left (keep all check-ins) adds M011's two rows (no member row) -> 7. Right (keep all members) adds M007 and M009 (no check-ins) -> 7. Outer adds all four -> 9.

Question 3 (SLO 04C)

```
checkins_long = pd.melt(
    checkins,
    id_vars=["member_id", "month"],
    value_vars=["week_1", "week_2", "week_3", "week_4"],
    var_name="week",
    value_name="visits",
)
```

member_id	month	week	visits
M008	Sep	week_1	7
M008	Sep	week_2	2
M008	Sep	week_3	3
M008	Sep	week_4	4
M008	Oct	week_1	5

member_id	month	week	visits
M008	Oct	week_2	3
M008	Oct	week_3	1
M008	Oct	week_4	3

8 rows: 2 wide rows x 4 week columns. Row order may differ (pd.melt actually lists all week_1 rows first, then week_2, …); any order is fine.

Question 4 (SLO 05A)

member	visits_per_week	calories (hundreds)	closest
M21	1	4	A
M22	2	5	A
M23	3	6	A
M24	4	1	B
M25	5	2	B
M26	6	3	B

New centroid A = (2, 5), new centroid B = (5, 2) – the mean visits_per_week and mean calories of each centroid's members.

Assign: each member goes to the centroid at the smallest straight-line distance. Update: each centroid moves to the average of its members.

Question 5 (SLO 05B)

k = 3. Inertia: the elbow is at 3 – it falls steeply from k = 1 to 3 (852 → 436 → 165), then only slowly (139, 116, 96, …). Silhouette: highest at k = 3 (0.61). Both agree. (Choosing k = 8 just because its inertia is lowest is wrong: inertia always drops as k grows.)

Question 6 (SLO 05C)

```
PCA().fit(X_scaled).explained_variance_ratio_.round(3)
# [0.617  0.307  0.070  0.006]
```

3 components: two keep 0.617 + 0.307 = 92.4% (< 95%); three keep 0.617 + 0.307 + 0.070 = **99.4%**.

Question 7 (SLO 03A) — Retake from Quiz 2

```
members["plan"] = (
    members["plan"]
    .str.strip()
    .str.lower()
    .str.replace(r"\s*-\s*", "-", regex=True)
)
```

member_id	plan (raw)	plan (after)
M006	`PREMIUM`	`premium`
M007	`Student-Monthly`	`student-monthly`
M008	`prem`	`prem`
M009	`Basic`	`basic`
M010	`student - monthly`	`student-monthly`

`\s*-\s*` replaces a hyphen and any spaces around it with a bare hyphen. Nothing turns `"prem"` into `"premium"`, so M008 would still not match `plans`.

Question 8 (SLO 03B) — Retake from Quiz 2

Answer:

```
result = members.groupby("plan").agg(
    n_members=("member_id", "count"),
    median_cardio=("cardio_pct", "median"),
)
```

Question 9 (SLO 03C) — Retake from Quiz 2

Question	Answer
What's the distribution of `avg_minutes` across all members?	Histogram – x=avg_minutes
How do `cardio_pct` and `calories` relate to each other?	Scatter – x=cardio_pct, y=calories
How does the average number of `visits` change from `week_1` to `week_4` (using `checkins_long`)?	Line – x=week, y=avg(visits)
How many members are in each `cluster` (after running k-means)?	Bar – x=cluster, y=count of members